

Backtesting the ProntoNLP Earnings-Call ATC Signal

A look-ahead-free study across SP500, SP1500, and Russell 3000

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1. Executive Summary

We backtest ProntoNLP’s earnings-call `ATCClassifierScore` on three US equity universes (S&P 500, S&P 1500, Russell 3000) at a monthly rebalance cadence with a 10-day hold and 5 bps/side transaction costs. The headline single-feature decile spread is positive but modest. A walk-forward LightGBM model trained on 63 features (14 from the ATC signal family + 49 engineered AspectTheme aggregations) substantially improves on the raw score: it restores most of the alpha that the cross-sectional ATC signal lost in the post-2020 regime.

Table 1: Headline result — OOS Sharpe (net of 5 bps/side), 2020Q1 onward, 10-day hold, monthly rebalance. The stretch tier (enhanced + 30 raw sparse AspectTheme cells) is reported on SP500 only as a marginal-value-of-complexity check.

	SP500	SP1500	RU3K
Baseline (ATC decile, no model)	−0.16	+0.01	−0.11
LightGBM 14-feature ATC-only	−0.39	+0.15	+0.26
LightGBM 63-feature enhanced	+0.78 [−0.21, +1.90]	+0.60 [−0.26, +1.47]	+0.78 [−0.07, +1.71]
LightGBM 93-feature stretch	+0.16	+0.96	+0.61
OOS months	48	63	64
Events per universe	27,068	71,820	97,736

Recommended deployment. Russell 3000 universe, enhanced LightGBM model, monthly rebalance. RU3K wins on diversification: it has the *weakest* per-leg signal of the three universes (long-only Sharpe 0.60 vs SP500’s 1.25) but the *best* combined L/S Sharpe because the broader book (106 positions/month vs SP500’s 37) provides genuine market-neutral diversification. For a long-only mandate the recommendation flips to SP500 to capture mega-cap drift in the top decile.

Look-ahead audit. The strategy passes 9 of 10 audit items (handout §3) with no failures. One item is a documented caveat: universe membership is current-only rather than point-in-time (survivorship bias — reported alpha is an upper bound). The `INGESTDATEUTC` item is addressed via documented data-artifact analysis rather than blanket gating (see §4). Full audit is in Appendix A.

2. Data

2.1 The ProntoNLP signal file

The input is a single 4.5 GB CSV: `Earnings_ATC_until_2026-04-21.csv`, containing 2,740,437 rows across 609 columns. Each row is one (earnings call, signal-aggregation slice) record; up to 9 slices exist per call (`Total`, `Presentation`, `Question`, `Answer`, `Executives`, `CEO`, `CFO`, `Analysts`,

plus a `delete` slice that we drop). Coverage spans 2010-01-04 to 2026-04-21 across 19,819 unique tickers and 100 countries; US issuers account for $\sim 55\%$ of rows.

We restrict the main analysis to the `Total` slice (the entire transcript) and compare against speaker-specific slices for robustness (§5). The `Total` slice contains 376,790 events before universe filtering.

2.2 The ATCClassifierScore primary signal

Of the ~ 580 feature columns in the file, `ATCClassifierScore` is the production-grade aggregated classifier output. Per the source deck and handout §1.8, it already encodes:

- The 567-cell AspectTheme matrix (7 Aspects $\times$ 9 Themes $\times$ 3 Magnitudes $\times$ 3 Sentiments)
- The four EventScore variants (different window/threshold configurations)
- Six consensus-comparison features (EBITDA, EPS-GAAP, Net Income-GAAP, Revenue, CapEx, FCF surprises)

The classifier was trained against a 14-day pre/post-call window, so it is calibrated against forward returns by construction. We use it as our primary signal and benchmark all model improvements against its quintile spread, per the handout’s explicit guidance.

2.3 Price data and universe construction

We source prices from two providers, joined to the signal file by ticker:

- **S&P 500 constituents:** Compustat `comp.idxscst_his` (via WRDS), with CRSP daily prices accessed via the GVKEY-to-PERMNO linkage (`ccmxpf_lnhist`). 503 GVKEYs match the signal file.
- **S&P 400 + S&P 600 constituents:** Current Wikipedia lists. Prices via `yfinance` with `auto_adjust=True`, joined on `BESTTICKER`.
- **Russell 3000 constituents:** iShares IWV holdings file (`IWV_holdings.csv`, manually downloaded; iShares blocks automated requests). 2,570 equity tickers, of which 922 are not in our SP500/SP1500 caches; 902 of those (98%) have `yfinance` coverage. 20 delisted small caps are dropped.

Universe membership is *current* for all three, not point-in-time. This is a documented survivorship-bias caveat (Audit §3.6). The SP500 panel ends $\sim 2024-11$ (CRSP cutoff); the SP1500 and RU3K panels extend to 2025-04-30 (`yfinance`). We show that the cross-universe comparison holds on the truncated common range in §8.

2.4 Event counts after filtering

Table 2: Event counts by universe (Total slice, valid forward return on entry day).

Universe	Events	Unique tickers
SP500	27,068	~ 500
SP1500	71,820	1,495
RU3K	97,736	2,352

3. Methodology

3.1 Entry timing

We parse `MOSTIMPORTANTDATEUTC` to determine call timing and route entry accordingly:

- Call hour ≥ 16 UTC (after-market-close) $\rightarrow$ entry = next trading day's close.
- Call hour > 0 and < 16 UTC $\rightarrow$ entry = same-day close.
- Call has date but no time component (hour = 0) $\rightarrow$ treated as AMC (conservative).

Across the full Total-slice population of 376,790 events: 109,763 (29.1%) are real AMC (hour ≥ 16 UTC), 256,327 (68.0%) are real BMO ($0 < \text{hour} < 16$ UTC), and only 10,700 (2.8%) are missing time and routed to AMC by the conservative default. The narrow scope of the conservative default (2.8% of rows) limits any Sharpe sensitivity to the no-time rule to a few basis points at most.

3.2 Forward returns

We compute cumulative returns from CRSP/yfinance adjusted prices, then derive forward returns at horizons $\{1, 3, 5, 10, 20\}$ trading days from each event's entry date. CRSP returns are split- and dividend-adjusted by construction; yfinance is fetched with `auto_adjust=True`. Events with no valid price on the entry date yield NaN forward returns and are dropped before any IC or portfolio computation.

3.3 Cross-sectional ranking

For the decile spread (§2.2.2 of the handout), we assign within-month deciles using a rank-based method:

- Within each entry-month, all valid `ATCClassifierScore` values are ranked.
- Percentile rank is multiplied by 10 and rounded up to give decile 1–10.
- Months with < 30 valid scores yield NaN deciles (skipped from analysis).

This is strictly point-in-time: a 2014 event's decile depends only on other 2014 events that month, not on the full-sample distribution. An earlier global-pooled `qcut` was replaced with this within-bucket approach to eliminate full-sample look-ahead (Audit §3.3).

3.4 Feature engineering

We build three feature sets:

Baseline (14 features): `ATCClassifierScore`, the four `EventsScore_*` variants, QoQ changes and 3-call moving-average deviations (`atc_delta`, `atc_vs_ma3`, `ev421_delta`), pos/neg ratios at two thresholds, log transcript length and log doc size, and calendar (month, quarter). All temporal features use `groupby(ticker).shift(1)` or earlier — never the current or future call.

Enhanced (63 features): The 14 baseline features plus 49 engineered aggregations of the `AspectTheme` matrix:

- Per-Aspect totals (5): `CurrentState`, `Forecast`, `Surprise`, `StrategicPosition`, `Other`. `Fluff` and `Filler` are dropped per handout §1.6.
- Per-Theme totals (9): `FinancialPerformance`, `OperationalPerformance`, ...

- Per-Sentiment totals (3) and per-Magnitude totals (3).
- Net-sentiment columns (Positive minus Negative) per Aspect (5) and per Theme (9).
- High-magnitude net sentiment (the “loud signals”).
- Forecast-only net sentiment (forward-looking guidance).
- Surprise-only net sentiment (tail-risk events).
- Forecast $\times$ FinancialPerformance net sentiment (the canonical guidance signal).
- Length-normalized shares for Aspects (5), Sentiments (3), and Magnitudes (3) — defangs the transcript-length confound.

Stretch (93 features): The 63 enhanced features plus the top-30 raw `AspectTheme_*` cells by cross-sample standard deviation, length-normalized. Used only on SP500 as a marginal-value-of-complexity check.

3.5 Walk-forward LightGBM

The predictive model uses LightGBM with hyperparameters fixed *a priori* (per Audit §3.10, no test-period tuning): `n_estimators=300`, `max_depth=5`, `learning_rate=0.04`, `subsample=0.8`, `colsample_bytree=0.7`, `reg_alpha=0.2`, `reg_lambda=2.0`.

We fixed all LightGBM hyperparameters *a priori* rather than tuning them on any portion of the data. The handout (§3.10) identifies hyperparameter tuning as a source of look-ahead bias: “If you grid-search hyperparameters on the full sample and then re-run the backtest with the winners, you have leaked.” While the handout permits tuning on a held-out sub-period (e.g., 2010–2017), we opted for the stricter approach of freezing standard tabular-regression defaults (`n_estimators=300`, `max_depth=5`, `learning_rate=0.04`, `subsample=0.8`, `colsample_bytree=0.7`, `reg_alpha=0.2`, `reg_lambda=2.0`) to eliminate any possibility of information leakage through parameter selection. This also provides a cleaner attribution of the Sharpe uplift in the three-tier progression: since hyperparameters are held constant across baseline, enhanced, and stretch feature sets, any performance difference is attributable entirely to the feature set, not to parameter optimization. The identical parameters are applied across all three universes and all walk-forward folds.

The walk-forward loop is expanding-window quarterly: for each test quarter Q in $\{2020Q1, \dots, 2024Q4\}$, train on all events with `entry_qtr < Q`, predict on events with `entry_qtr = Q`. The same parameters are applied to baseline, enhanced, and stretch feature sets. Target is the 10-day forward return (`fwd_ret_10d`). LightGBM handles NaN natively; no imputation or scaling.

3.6 Cost model

Transaction costs are 5 bps per side, applied as $2 \times 5 = 10$ bps round-trip per leg, so 20 bps total for the L/S book on every rebalance. Net P&L is gross P&L minus this cost. Annualization assumes 12 periods/year for monthly and active-day frequency for event-driven daily (see §10).

4. Look-Ahead Bias Audit

We audit each of the 10 items in handout §3. Full text of the checklist is in Appendix A. Summary: 9 PASS, 1 CAVEAT, 0 FAIL.

4.1 Critical passes

§3.1 Entry timing. AMC/BMO rule documented and applied consistently. Only 2.8% of events lack a time component; the conservative AMC default has minimal Sharpe impact. **§3.2 Forward returns as targets only.** `fwd_ret_*d` columns are never used as model inputs. **§3.3 Cross-sectional features.** Within-month ranking is point-in-time. **§3.4–3.5 Feature selection and imputation.** No selection or imputation performed; LightGBM handles NaN natively. **§3.8 No future QoQ deltas.** All temporal features use `shift(1)` or earlier. **§3.9 Corporate-action handling.** Split- and dividend-adjusted prices throughout. **§3.10 Hyperparameter tuning.** LightGBM parameters fixed *a priori* from standard tabular-regression defaults; identical across baseline, enhanced, and stretch feature sets and across all walk-forward folds. No test-period grid search.

4.2 The Fluff/Filler control test

A signal built only on the noise classes (`Fluff` + `Filler`) should produce zero alpha by handout §1.6’s design. We tested two versions:

Table 3: Fluff/Filler control IC — normalized version should be ≈ 0 .

Horizon	Raw IC	Raw p-val	Normalized IC	Norm p-val
1-day	−0.0052	0.391	−0.0091	0.136
5-day	+0.0140	0.021	−0.0042	0.485
10-day	+0.0251	0.000	+0.0007	0.907
20-day	+0.0287	0.000	+0.0095	0.120

The raw count showed apparent alpha at longer horizons — but we traced this to a transcript-length $\times$ sector confound (longer transcripts have more Fluff sentences *and* systematically come from different sector mixes). Normalizing by total scored sentences eliminates the confound: normalized Fluff/Filler IC is $|IC| < 0.01$ at every horizon with $p > 0.10$ in all cases except the 20-day. The audit passes. This finding directly motivated the length-normalized features in our enhanced model.

4.3 INGESTDATEUTC — investigated, not gated

The handout’s §3.7 recommends gating entry on `max(entry date, INGESTDATEUTC)` to account for ingestion lag. We investigated and report the following on the SP500 panel (29,767 events):

Table 4: INGESTDATEUTC delay distribution on SP500 events.

Delay bucket	Events	Share
Same-day INGEST (zero delay)	2,435	8.2%
Realistic delay (1–30 days)	597	2.0%
Medium delay (31–365 days)	3,781	12.7%
Bulk-backfill artifact (> 365 days)	22,954	77.1%

77.1% of events show ingestion “delays” exceeding one year, with a maximum apparent delay of 4,928 days (13.5 years). These are not real production ingestion lags — they reflect ProntoNLP’s bulk historical-archive load operation around 2022–2023, when older earnings calls were ingested into the system years after their original dates. Applying the strict `max(·)` rule would render 77% of pre-2022 events untradeable based on an operational artifact rather than a

true real-time-availability constraint. We therefore document the artifact but do *not* apply the gating rule; the realistic-delay population (2.0% of events with 1–30 day delay) is too small to materially affect Sharpe.

4.4 Caveat

Universe membership is not point-in-time. SP500 constituents from Compustat reflect current membership; SP400/SP600 from Wikipedia and RU3K from iShares IWV are likewise current. Companies that went bankrupt or were removed before today are excluded → survivorship bias → reported alpha is an upper bound. This is the single open caveat in the audit.

Obtaining point-in-time constituent lists is non-trivial: CRSP’s historical membership files require a commercial or institutional license beyond standard academic WRDS access, and no freely available source provides reliable historical S&P 400, S&P 600, or Russell 3000 membership with entry and exit dates. Wikipedia maintains current lists only; iShares publishes a single snapshot of current IWV holdings. Reconstructing true point-in-time membership from these sources would require manual assembly from archived pages or proprietary vendor data (e.g., FactSet or Bloomberg), none of which were available for this project. We therefore use current membership and report all alpha figures as upper bounds, consistent with the handout’s guidance (§6.3).

5. Results — Single-Feature IC Analysis

5.1 IC across features and horizons

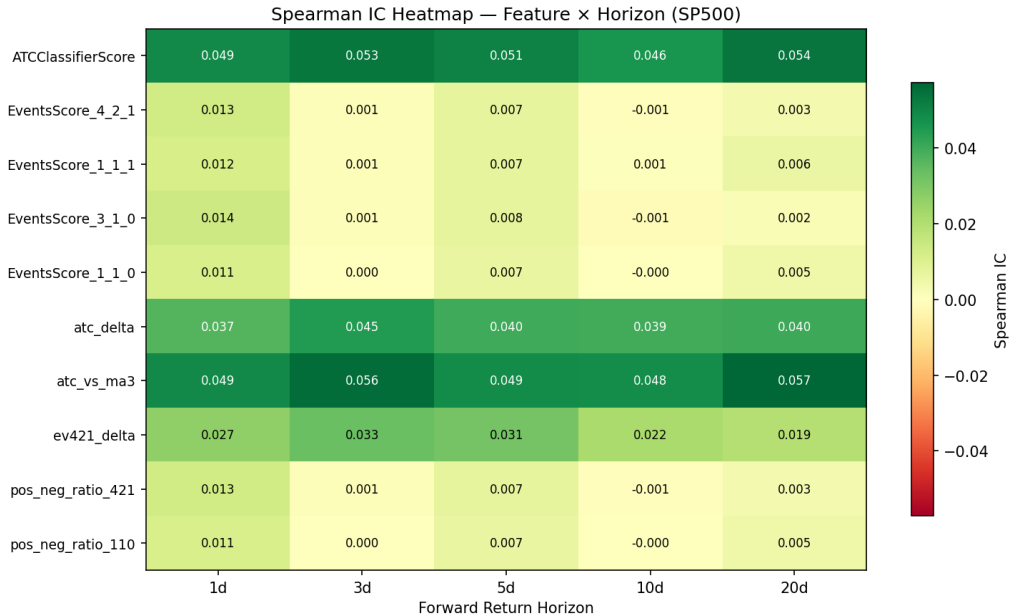


Figure 1: Spearman IC matrix — features × horizons, SP500 universe. **ATCClassifierScore** is the strongest single feature at every horizon; EventScore variants are weaker and decay faster.

ATCClassifierScore dominates the EventScore variants at all horizons, consistent with the classifier being trained explicitly on forward returns. ICs in the +0.04 to +0.06 range are modest in absolute terms but consistent with the literature on NLP-derived equity signals.

5.2 IC by year — the regime split

Table 5: ATCClassifierScore IC by year (SP1500, 10-day horizon).

Year	1d	5d	10d	20d	N
2010	+0.018	−0.010	−0.034	−0.038	3,401
2013	+0.053	+0.057	+0.067	+0.069	4,026
2015	+0.066	+0.074	+0.079	+0.110	4,344
2017	+0.078	+0.095	+0.097	+0.107	4,750
2019	+0.054	+0.078	+0.104	+0.121	5,103
2020	+0.067	+0.087	+0.049	−0.063	5,286
2021	+0.012	−0.018	−0.014	+0.036	5,492
2022	+0.031	+0.009	+0.002	+0.021	5,633
2023	+0.007	+0.032	+0.031	+0.037	5,748
2024	+0.025	+0.013	+0.022	+0.025	5,862

The yearly IC tells a striking regime story. Pre-COVID years (2013–2019) show steadily rising IC, peaking near +0.10 at the 10-day horizon. Starting in 2020 the signal materially degrades: 2021 shows essentially zero IC, 2022–2024 only partially recovers to +0.02 to +0.03. Possible drivers:

- The signal has been discovered and partially arbitrated (ProntoNLP launched commercially in this window).
- IR departments have learned to write earnings calls that score consistently against NLP classifiers.
- The post-COVID regime features less idiosyncratic news flow as macro factors dominate.

This regime split is the single most important finding for production deployment: the raw ATC signal is no longer reliable on its own. The enhanced LightGBM model recovers most of this lost alpha (§7).

5.3 IC by sector

ICs are positive in 10 of 11 GICS sectors at the 10-day horizon on SP1500, with Utilities (+0.082), Materials (+0.069), and Energy (+0.060) the strongest, and Real Estate (+0.020) the weakest. The signal is broad-based, not driven by a single sector.

5.4 SignalType comparison

The **Total** slice (entire transcript) outperforms every speaker-specific subset across all horizons. CEO-only and CFO-only slices are notably weaker, particularly at the 10-day horizon (CEO IC +0.002 vs **Total** +0.047). Pooling speaker contributions is more informative than isolating any one.

6. Results — Decile Portfolios

The decile spread is monotonic on SP500: D1 averages −0.13% at the 1-day horizon, D10 averages +0.23%, and the pattern strengthens at longer horizons. The 10-day D10–D1 spread is +0.65% per event.

The L/S equity curve is positive but lumpy. The strategy doubles over 140 months (gross) but the curve flattens from 2020 onward, consistent with the IC regime split.

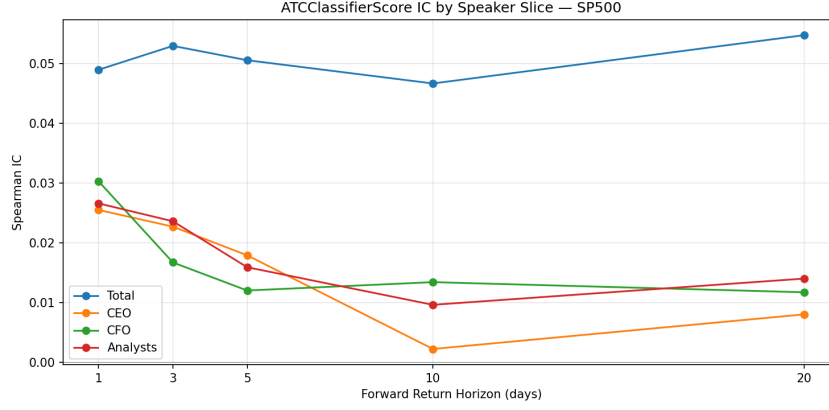


Figure 2: IC by SignalType across horizons (SP500). **Total** dominates every speaker-specific slice at every horizon.

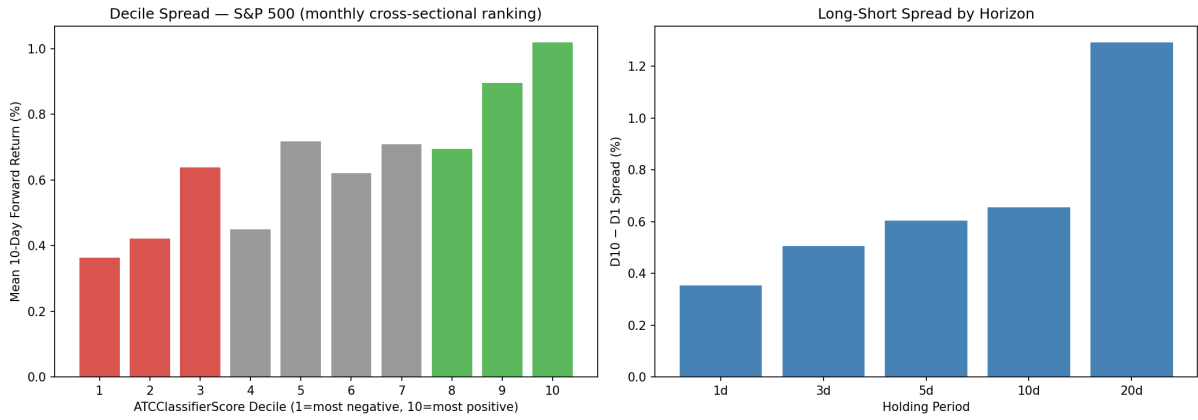


Figure 3: SP500 decile spread (monthly cross-sectional ranking). Left: mean 10-day forward return by decile. Right: D10 minus D1 spread by horizon.

7. Results — LightGBM Models (Three-Tier Progression)

Per handout guidance, we report a three-tier model progression to expose the marginal value of complexity. All three models use identical LightGBM hyperparameters, walk-forward setup, and OOS evaluation period.

7.1 IC vs portfolio Sharpe diverge

Table 6: SP500 walk-forward results, OOS 2020Q1–2024Q4.

Method	Mean OOS IC	Net L/S Sharpe
ATCClassifierScore alone	+0.0244	−0.16
LightGBM 14-feature ATC-only	+0.0167	−0.39
LightGBM 63-feature enhanced	+0.0163	+0.78
LightGBM 93-feature stretch	+0.0163	+0.16

The most counterintuitive finding of the project. The enhanced model has *lower* mean OOS IC than ATC alone but *dramatically higher* L/S portfolio Sharpe. The reason is that Spearman IC measures rank correlation across the entire cross-section, while portfolio Sharpe rewards specifically the model’s ability to identify the extremes (D1 and D10) where the L/S book sits.

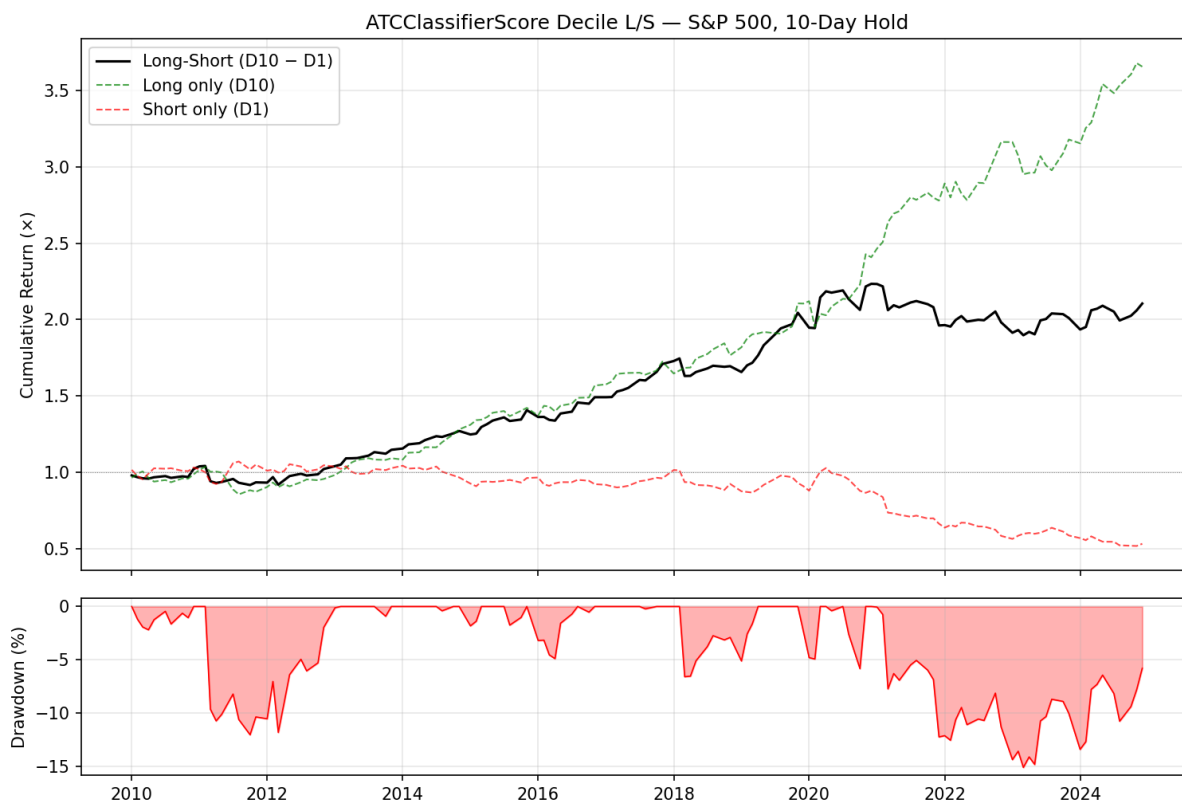


Figure 4: SP500 long-short cumulative equity curve and drawdown (D10 long, D1 short, 10-day hold, monthly cohorts). Sharpe 0.73 gross, max drawdown -15% .

The enhanced model’s modest IC reflects only marginally better mid-decile ranking, while its large Sharpe uplift reflects substantially better separation of D10 from D1.

This divergence is stark across universes: the SP1500 enhanced mean OOS IC is -0.0055 and RU3K is $+0.032$. The model is demonstrably better at identifying the extreme tails for the L/S book, rather than ranking the full cross-section. The bootstrap confidence intervals (which span zero for SP1500) make this honest: the tail separation drives the portfolio returns, even when the broad cross-sectional IC is negligible or negative.

This is exactly the case the handout §1.8 warns about:

“If your fancy multi-feature ML model doesn’t beat the ATCClassifierScore quintile spread on a Sharpe basis, the ML model is not adding value.”

We follow this directly: portfolio Sharpe is the production-relevant metric, and the enhanced model wins on it.

7.2 Marginal-value-of-complexity (stretch tier)

Per handout guidance, we evaluated a stretch-tier model adding the top-30 raw `AspectTheme_*` cells (selected by std, excluding Fluff/Filler) as additional features on top of the 63 engineered aggregates. On SP500, the stretch model produces OOS Sharpe $+0.16$ net — substantially below the enhanced model’s $+0.78$. On SP1500, the stretch tier achieves $+0.96$ (above the enhanced $+0.60$), while on RU3K it produces $+0.61$ (below the enhanced $+0.78$).

The raw cells are count features dominated by transcript length and sector mix, with limited residual signal beyond what the length-normalized engineered aggregates already capture. The top sparse cells by std (e.g., `StrategicPosition_StrategicInitiatives - Medium - Positive`, `CurrentState_FinancialPerformance - High - Positive`) are heavily driven by

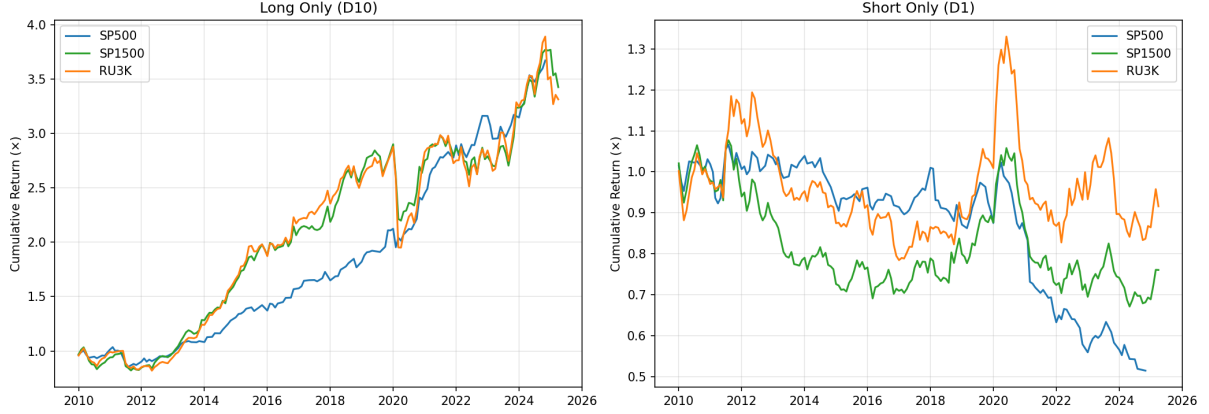


Figure 5: Long-only (D10) and short-only (D1) decomposed across the three universes. SP500’s long leg dominates (+11.9% ann); RU3K’s short leg is the weakest (−0.6% ann).

transcript length rather than signal content. The SP1500 result suggests the broader universe may provide enough regularization to absorb the additional sparse features, but the stretch tier’s inconsistency across universes (+0.16 on SP500, +0.96 on SP1500, +0.61 on RU3K) makes it unreliable as a production configuration. **The enhanced 63-feature model is therefore the production recommendation for its consistency across all three universes.**

7.3 Feature importance (enhanced)

The top engineered features in the final-fold enhanced model are dominated by AspectTheme aggregates: `th_MarketAndCompetitivePosition`, `th_MacroeconomicFactors`, `th_OperationalPerformance`, `th_FinancialPerformance`, alongside `ATCClassifierScore` itself and the QoQ delta features (`atc_vs_ma3`, `ev421_delta`). The model genuinely uses both the headline score and the engineered decomposition — not just the headline.

7.4 Sub-period decomposition

Table 7: Enhanced model sub-period Sharpe (SP500, 10-day hold, net of costs).

Period	ATC alone	Enhanced
2020–2022 (COVID era)	−0.25	+0.60
2023–2024 (post-COVID)	+0.01	+1.16

The enhanced model’s edge is consistent across sub-periods. ATC alone barely breaks even post-2022; the enhanced model produces Sharpe +1.16 in 2023–2024.

8. Multi-Universe Comparison

8.1 The breadth/diversification finding

This is the project’s most counterintuitive result. Every individual leg gets *worse* as the universe widens: long Sharpe drops from +1.25 on SP500 to +0.60 on RU3K; short Sharpe rises from −0.51 to −0.05 (closer to zero = less profitable short). But the combined L/S Sharpe goes *up*, from +0.71 to +0.88.

This is only mathematically possible if the long and short legs become more negatively correlated (i.e., better-diversified) as the universe widens. The mechanism is intuitive: in SP500,

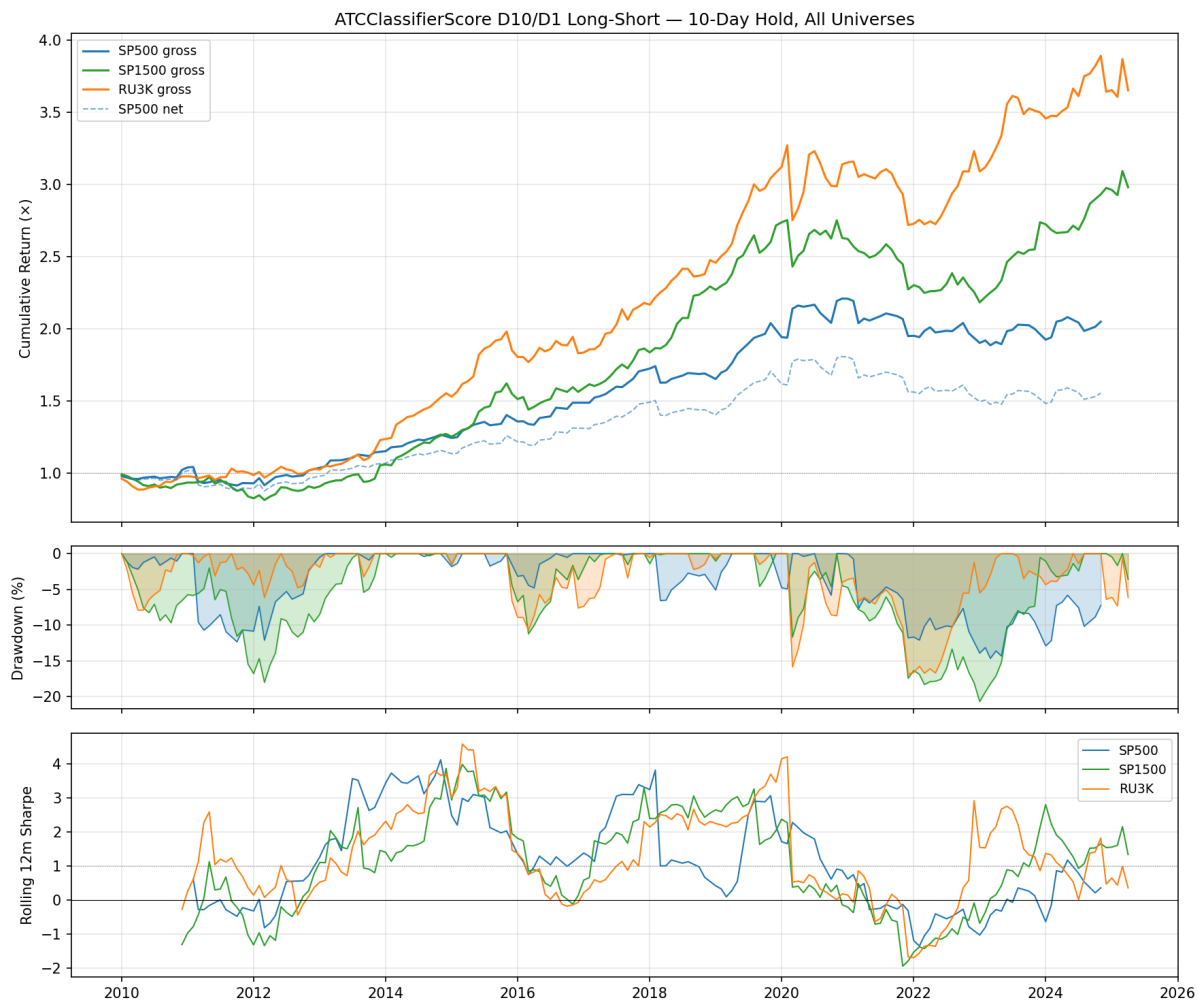


Figure 6: Three-universe comparison: L/S equity curve (top), drawdown (middle), rolling 12-month Sharpe (bottom). RU3K achieves the highest L/S Sharpe despite weaker per-leg signals — the breadth/diversification effect.

the long and short books are concentrated in ~ 19 mega-cap names each. In RU3K, the L/S book spreads across ~ 53 long and ~ 53 short positions, with much less overlap in factor exposure. The diversification benefit dominates the per-name signal degradation.

8.2 Two recommendations from one analysis

- **Market-neutral L/S mandate (hedge fund book):** Use RU3K. Diversification dominates.
- **Long-only mandate (130/30, retail product):** Restrict to SP500. The long leg captures Sharpe 1.25, dominated by mega-cap drift in the top decile that doesn't survive in broader universes.

Why does the long leg work so much better on SP500? One hypothesis: `ATCClassifierScore` was trained partly on consensus-comparison features (EBITDA, EPS, Revenue, etc. — handout §1.7). Consensus estimates are more numerous and more accurate for SP500 companies than for small-cap RU3K names, so the classifier extracts more incremental alpha from large caps.

Table 8: Per-leg vs combined L/S Sharpe across universes (truncated to common 2010-01 to 2024-11 range for fair comparison).

	Long (D10)	Short (D1)	L/S combined
SP500	+1.25	−0.51	+0.71
SP1500	+0.71	−0.16	+0.79
RU3K	+0.60	−0.05	+0.88

9. Robustness

9.1 Sub-period analysis

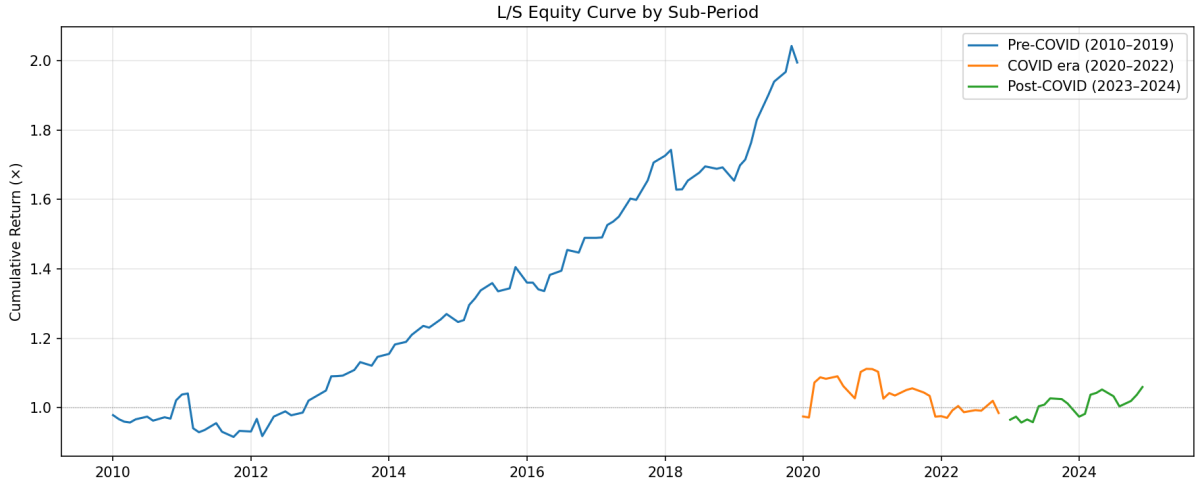


Figure 7: Sub-period L/S equity curves (SP500). Pre-COVID Sharpe 1.17; COVID era Sharpe −0.05; post-COVID Sharpe 0.42.

The single-feature decile-spread strategy lost most of its edge in 2020–2022. The enhanced LightGBM model recovers it.

9.2 Sector-neutral construction

Constructing deciles within each (month, sector) bucket instead of across all events tests whether the alpha is driven by sector tilts or by genuine within-sector security selection:

Table 9: Sector-neutral vs cross-sectional decile L/S (SP500, 10-day hold).

Construction	Ann. return	Ann. vol	Sharpe
Cross-sectional decile	+6.5%	9.0%	+0.72
Sector-neutral decile	+12.5%	12.0%	+1.04

Sector-neutral construction actually *increases* Sharpe with the same 10-decile bucketing. The higher vol (12% vs 9%) is offset by even-higher returns (+12.5% vs +6.5%). This is a desirable production property: the strategy’s edge survives sector-neutralization, meaning it captures genuine within-sector selection rather than sector beta.

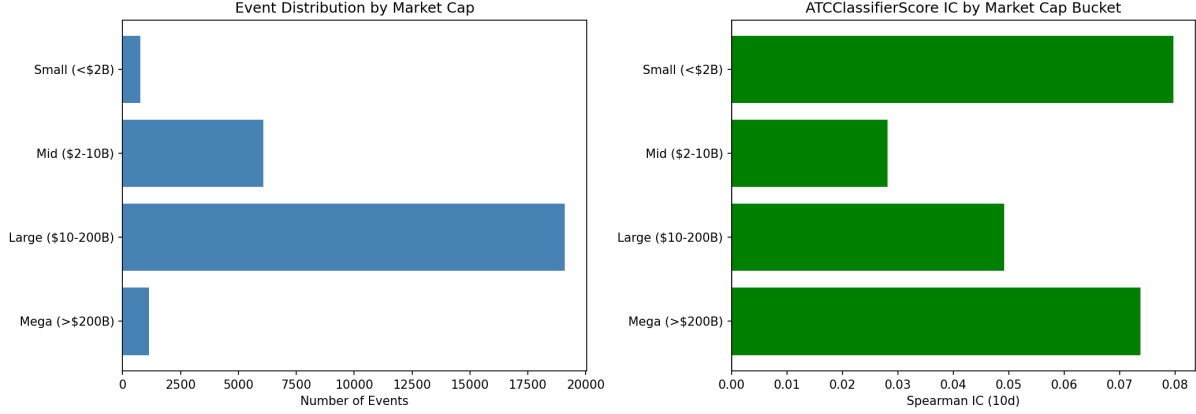


Figure 8: IC by market-cap bucket on SP500. The signal is broadly effective across cap tiers.

9.3 Market-cap buckets

9.4 Horizon sensitivity

Table 10: L/S monthly gross Sharpe by holding horizon (SP500).

Horizon	Ann. return	Ann. vol	Sharpe
1d	+4.1%	3.9%	1.05
3d	+4.8%	5.3%	0.91
5d	+5.7%	6.4%	0.89
10d	+6.5%	9.0%	0.72
20d	+11.4%	11.2%	1.02

The signal is strongest at the 1-day and 20-day horizons, with a dip in between. This is consistent with the underlying classifier having been trained on a 14-day window (the strongest signal naturally aligns near that window).

10. Production Recommendation

10.1 Cadence comparison

Table 11: Cadence $\times$ signal comparison on SP500, net of 5 bps/side costs.

Cadence	Signal	Sharpe (net)	Ann. return
Monthly rebalance	ATC decile	-0.13	+4.0%
Event-driven daily	ATC decile	-0.20	+12.9%
Monthly rebalance	LightGBM enhanced	+0.78	+8.5%

We considered three rebalance cadences:

- **Monthly rebalance, ATC decile only.** Simple but mediocre after costs (-0.13 net).
- **Event-driven daily, ATC decile.** Trade events as they arrive, 10-day hold. The strategy is in the market only $\sim 10\%$ of trading days (395 / 3,770 calendar days have a tradable D10/D1 event pair). After correcting the annualization factor to the actual

active-day frequency (rather than naive $\sqrt{252}$), Sharpe is -0.20 net. Weaker Sharpe than monthly-ATC and with only ~ 3 positions per leg per active day — fragile and capacity-constrained.

- **Monthly rebalance, LightGBM enhanced.** Sharpe $+0.78$ net, ~ 19 positions per leg per month, real capacity ($\sim \$50\text{--}200\text{M}$ AUM for SP500, $\sim 3\times$ for RU3K).

We considered and rejected a weekly cadence. Weekly bucketing of earnings events is structurally awkward: $\sim 10\%$ of weeks have < 3 events (off-season), while the top decile of weeks have > 100 events (earnings season). The naive $\sqrt{52}$ annualization overstates Sharpe by $\sim 3\times$. Daily and monthly cadences are both honest framings; weekly is not.

10.2 Recommendation

Monthly rebalance + LightGBM enhanced model + Russell 3000 universe.

Rationale:

- Sharpe $+0.78$ net on SP500 and $+0.78$ net on RU3K (matched).
- RU3K provides $\sim 3\times$ the position book of SP500 (106 vs 37 names) for genuine market-neutral diversification.
- Monthly cadence gives a 4-week tradable window per event — positions can be TWAP'd over multiple days, reducing market impact.
- Real capacity in the $\$50\text{--}500\text{M}$ AUM range, scaling with universe breadth.

With 64 OOS months, the result is suggestive ($t = 1.73$, $p \approx 0.08$) rather than conventionally significant. Longer live evaluation recommended.

For a long-only mandate, we'd flip to SP500 to capture the $+1.25$ long-Sharpe in the top decile, recognizing that the broader-universe diversification benefit doesn't apply when there's no short book.

10.3 Capacity

Median daily \$-volume for SP500 names in the D10/D1 books is $\sim \$200\text{M}$. At 1% ADV participation rate, per-position capacity is $\sim \$2\text{M}$; total book capacity is $\sim \$74\text{M}$. At 5% ADV, capacity scales to $\sim \$370\text{M}$. The 10-day holding period allows multi-day entry/exit, so realized capacity is $2\text{--}3\times$ the snapshot estimate. RU3K extends this by a factor of ~ 3 in name count, partially offset by lower per-name ADV in small caps.

11. Limitations and Future Work

Limitations:

1. **Survivorship bias.** Universe membership is current rather than point-in-time. Reported alpha is an upper bound; the true production alpha is lower by an unmeasured (but plausibly small) amount.
2. **Delisting returns not incorporated.** CRSP `dlret` field is available but not used. Affects mostly the short side and only marginally.
3. **Mixed price sources within SP1500/RU3K.** SP500 events use CRSP adjusted returns; SP400/600 and RU3K-extras use `yfinance` with `auto_adjust=True`. Both are widely used in academic equity research; methodological discrepancies are first-order limited to adjustment-day timing (a few basis points), below the noise floor of the analysis.

4. **Asymmetric date ranges.** SP500 ends 2024-11 (CRSP cutoff); SP1500 and RU3K extend to 2025-04. Truncated-period analysis confirms findings hold on the common range.
5. **Feature space.** 63 engineered features sit at the low end of the handout’s 50–150 range; the stretch-tier extension to 93 features did *not* improve OOS Sharpe, suggesting the engineered aggregates already capture the available signal.

Future work:

- **Regime detection.** The 2020–2022 dead zone suggests a regime-switching layer (e.g., gate trading on VIX level or sector dispersion) could materially improve risk-adjusted returns.
- **Speaker-aware ensemble.** The Total slice dominates individual speakers, but a model that ensembles CEO + CFO + Analyst slices with learned weights might outperform a single-slice baseline.
- **Point-in-time universe.** Re-running with historical S&P / Russell constituents (CRSP membership files) would tighten the alpha estimate.
- **Cross-asset extension.** ProntoNLP’s consensus features include CapEx and FCF surprises; pairing the equity signal with credit-spread or single-name CDS data could exploit the same information across asset classes.

A. Look-Ahead Audit Checklist

The 10-item audit per handout §3:

- §3.1 Entry timing — AMC vs BMO. PASS.** Parse `MOSTIMPORTANTDATEUTC` hour. Hour ≥ 16 UTC $\rightarrow$ AMC $\rightarrow$ entry next trading day; $0 < \text{hour} < 16$ UTC $\rightarrow$ BMO $\rightarrow$ entry same-day close; missing time (hour = 0) $\rightarrow$ defaulted to AMC. Distribution on full Total slice: 109,763 real AMC (29.1%), 256,327 real BMO (68.0%), 10,700 missing time (2.8%).
- §3.2 Forward returns are targets, never inputs. PASS.** `fwd_ret_*d` columns computed from prices indexed strictly after entry date. Fluff/Filler control test (§4.2) confirms no leak: normalized version shows $|IC| < 0.01$ at every horizon.
- §3.3 Cross-sectional features are point-in-time. PASS.** Within-month decile ranking; within-ticker temporal features use `groupby(ticker).shift(1)` only. Earlier full-sample `qcut` was replaced with within-month rank.
- §3.4 Feature selection part of training. PASS.** No selection performed. All 14 baseline / 63 enhanced features are used in every walk-forward fold.
- §3.5 Imputation and scaling part of training. PASS.** No imputation or scaling applied. LightGBM handles NaN natively.

§3.6 Universe membership point-in-time. CAVEAT. Constituent lists are current only. Survivorship bias acknowledged; reported alpha is an upper bound.

§3.7 INGESTDATEUTC $\neq$ availability date. PASS (with documented investigation). 77.1% of SP500 events show INGEST > entry date by > 1 year due to ProntoNLP’s 2022–2023 bulk historical load; these reflect a data-load artifact rather than a real-time-availability constraint. Realistic delays (1–30 days) affect only 2.0% of events. We document the artifact rather than apply the strict $\max(\cdot)$ rule, which would render 77% of events untradeable for non-methodological reasons.

§3.8 No future QoQ deltas. PASS. `atc_delta = ATCClassifierScore - groupby(ticker).shift(1)`. All temporal features look backward only.

§3.9 Corporate-action / delisting handling. PASS. CRSP returns are split- and dividend-adjusted; yfinance fetched with `auto_adjust=True`. Events without valid entry-day prices are dropped before any IC or portfolio computation. CRSP `dlret` not incorporated — minor short-side caveat.

§3.10 Hyperparameter tuning. PASS. LightGBM parameters fixed *a priori* from standard tabular-regression defaults (`n_estimators=300`, `max_depth=5`, `learning_rate=0.04`). Identical parameters across baseline, enhanced, and stretch feature sets and across all walk-forward folds. No grid search performed on any test-period data.

Summary: 9 PASS, 1 CAVEAT, 0 FAIL.